LogicCircuit: Building Digital Circuits Made Easy

Open source software LogicCircuit is a handy assistant for digital circuit and logic gate design. Originally intended for educational purposes, this free software has been expanded to medium-level practical use as well. It is used to design and simulate digital logic circuits.

The LogicCircuit interface is a blend of drag-and-drop actions and symbolic features based on a simple layout. It has two functional modes: edit mode, where you can create and edit circuits, and run mode, where you can simulate the functioning of the designed circuit. The software packs an array of functionalities that cover the essentials of a logic circuit design. We look into all these features and how they make a compact design and analysis tool for circuit developers.

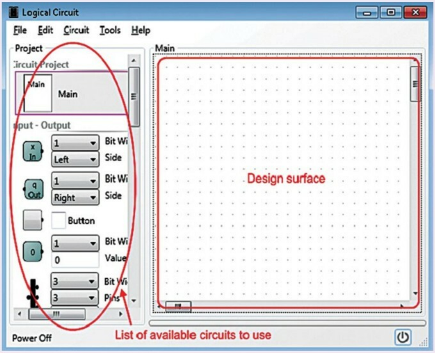

Fig. 1: The LogicCircuit interface (Image source: http://www.logiccircuit.org)

Creating and editing circuits

Edit mode of the software opens with a workspace (design surface) in the middle, a toolbar with multiple options on the top and a left panel. The left panel contains all the components and parameters that can be incorporated in a circuit, while the central workspace starts up blank, where the main designing and editing can be done. Each desired component can be dragged and placed on the design surface from the panel.

Components can be moved around within the surface with a similar dragging function. The wiring in the circuit can be indicated by connecting the output point of one component to that of another with the help of your mouse. A group of elements can also be moved around together by

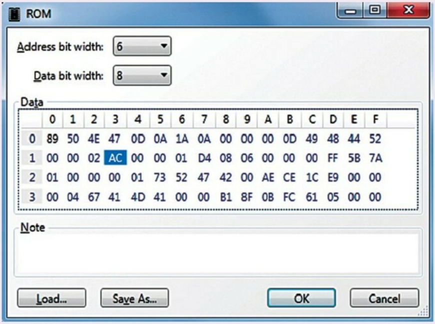

Fig. 2: ROM editor